GEOPHYSICAL SURVEY REPORT

# STRATASCAN™

SUMO Survey Services

SUMO Group Member

Project name:

## Land East of Swindon

Client:

## CGMS

## May 2013

Job ref:

**J3305**

Report author:

**Dr John Gater MiFA FSA**

# GEOPHYSICAL SURVEY REPORT

Project name:
**LAND EAST OF SWINDON**
Client:
**CgMs**

Job ref:
**J3305**

Techniques:
**Detailed magnetic survey – Gradiometry**

Survey date:
**[18 - 28 March 2013]**

Site centred at:
**SU 200 870**

Post code:
**SN3 4SH**

Field team:
**Jimmy Adcock, Graeme Attwood, Finn Pope-Carter, Dan Shiel, Claire Stephens, Emma Watson, Emma Wood**

Project manager:
**Simon Haddrell** BEng(Hons) AMBCS PIFA

Report written By:
**Dr John Gater** MiFA FSA

CAD illustrations by:
**Emma Wood** BSc MiFA , **Graeme Attwood** MSc

Checked by:
**David Elks** MSc AIFA

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# 1 SUMMARY OF RESULTS

A detailed gradiometer survey was conducted over approximately 35 hectares of mixed agricultural land located to the northeast of Swindon, immediately south of South Marston. The data have identified a multitude of anomalies that have been highlighted as being either of *probable* or *possible* archaeological origin. These comprise a series of ditched enclosures, some forming presumed paddocks or settlement enclosures, including so-called 'ladder type' settlement and others indicating sub-rectangular and circular / ring features. The high density and nature of the responses suggest multi-period activity. The archaeological responses appear to fall within two zones: one in the Southwest and the other in the northeast of the survey area.

Many of the fields simply indicate modern agricultural activity, whilst magnetic disturbance is evident throughout the survey areas.

# 2 INTRODUCTION

## *2.1 Background synopsis*

Stratascan were commissioned to undertake a geophysical survey of an area outlined for development. This survey forms part of an archaeological investigation being undertaken by CgMs.

## *2.2 Site location*

The site is located northeast of Swindon at OS ref. SU 200 870

## *2.3 Description of site*

The survey area consists of *c.* 60 hectares, approximately 35 hectares was surveyed and consisted of land presently used for turf cutting; pasture and cereal production. Some areas were classified as unsurveyable due to the presence of buildings, horses and associated fencing. Some areas were underwater / totally waterlogged which also precluded survey. Approximately 25 hectares were not surveyed due to deeply ploughed soils and access denial on a turf field.

## *2.4 Geology and soils*

The underlying geology is partly Ampthill and Kimmeridge Clay Formation (Mudstone) and partly Red Down Sand Member (Ferruginous Sandstone). The drift geology is unrecorded. (British Geological Survey website).

The overlying soils are either from the Foggathorpe association (712hh) and comprise slowly permeable seasonally waterlogged clayey and fine loamy soils or from the Burlesden association (572j) which are coarse loamy soils which are seasonally waterlogged and affected

by groundwater (Soil Survey of England and Wales, 1983).

## 2.5 *Site history and archaeological potential*

Past geophysical work at the site has identified a complex of archaeological features spread across a wide zone.

## 2.6 *Survey objectives*

The objective of the survey was to locate any features of possible archaeological origin in order that they may be assessed prior to development.

## 2.7 *Survey methods*

This report and all fieldwork have been conducted in accordance with both the English Heritage guidelines outlined in the document: *Geophysical Survey in Archaeological Field Evaluation, 2008* and with the Institute for Archaeologists document *Standards and Guidance for Archaeological Geophysical Survey.*

Detailed magnetic survey (gradiometer) was used as an efficient and effective method of locating archaeological anomalies. More information regarding this technique is included in the Methodology section below and in Appendix A.

## 2.8 *Processing, presentation and interpretation of results*

### 2.8.1 *Processing*

Processing is performed using specialist software. This can emphasise various aspects contained within the data but which are often not easily seen in the raw data. Basic processing of the magnetic data involves 'flattening' the background levels with respect to adjacent traverses and adjacent grids. Once the basic processing has flattened the background it is then possible to carry out further processing which may include low pass filtering to reduce 'noise' in the data and hence emphasise the archaeological or man-made anomalies.

The following schedule shows the basic processing carried out on all minimally processed gradiometer data used in this report:

1. *Destripe* (Removes striping effects caused by zero-point discrepancies between different sensors and walking directions)
2. *Destagger* (Removes zigzag effects caused by inconsistent walking speeds on sloping, uneven or overgrown terrain)

### 2.8.2 *Presentation of results and interpretation*

The presentation of the data for each site involves a print-out of the minimally processed data both as a greyscale plot (Figures 3,6,9) and a colour plot showing extreme magnetic values (Figures 2,5,8). Magnetic anomalies have been identified and plotted onto the 'Abstraction

and Interpretation of Anomalies' drawing for the site (Figures 4,7,10).

# 3 RESULTS

The detailed magnetic gradiometer survey conducted at South Marston, northeast of Swindon identified a number of anomalies that have been characterised as being either of a *probable* or *possible* archaeological origin.

The difference between *probable* and *possible* archaeological origin is a confidence rating. Features identified within the dataset that form recognisable archaeological patterns have been interpreted as being of a probable archaeological origin.

Features of possible archaeological origin tend to be more amorphous anomalies which may have similar magnetic attributes in terms of strength or polarity but are difficult to classify as being archaeological or natural.

The following list of numbered anomalies refers to numerical labels on the interpretation plots (Figures 4,7,10).

## *3.1 Probable Archaeology*

Archaeological-type responses have been found in two broad zones within the development area.

**1-5** Within Fields 1, 5, 6, 9 and 22, a complex of magnetic anomalies has been recorded. The features identified comprise presumed settlement enclosures, (including in Field 1 a 'ladder-type' settlement, stock enclosures, field systems and both rectilinear and circular features. There appear to be numerous pit-like features. The complex of features is very extensive and suggests multi-period activity.

**6 -8** In Field 15 a separate concentration of anomalies, that are also almost certainly archaeological in origin, has been identified. This concentration comprises an area of generally enhanced magnetic responses, typical of those near to settlement sites where there tends to be increased magnetic enhancement, and several distinct features. The latter include at least three ring ditches (**6**), a rectangular enclosure (**7**) plus linear ditches and a possible trackway (**8**).

## *3.2 Possible Archaeology*

Less clearly defined responses have been classified as *Possible Archaeology*; these may include plough-damaged or less magnetically enhanced features.

A former field boundary within Field 15 is marked on 1st edition Ordnance Survey maps; this boundary also marks the limit of the magnetic enhancement as mentioned in paragraph 3.2 above.

### *3.3 Other Anomalies*

Areas of magnetic disturbance are the result of nearby ferrous metal objects such as fences and underground services. These effects can mask weaker archaeological anomalies, and on this site may have affected the results on the periphery of the modern fields.

A number of magnetic 'spikes' indicate ferrous metal objects; these are likely to be modern debris.

Closely spaced, parallel linear responses are noted across many of the fields and are indicative of modern agricultural activity, usually ploughing.

## 4 CONCLUSION

The gradiometer data were collected from a number of fields covering approximately 35 hectares of land to the Northeast of Swindon. The results indicate a complex of archaeological features associated with settlement and stock enclosures; former field systems; three well-defined ring ditches (and other less clear examples); several rectilinear enclosures; a wealth of pits and some old trackways of presumed antiquity.

When viewed in conjunction with earlier geophysical results, there appear to be two main archaeological complexes, one in the southwest of the survey area and a second in the northeast. Multi-period activity seems highly likely.

Several fields within the proposed development area appear to be devoid of archaeological-type responses, certainly of a character similar to that identified above.

# 5 REFERENCES

British Geological Survey, n.d., *website*: (http://mapapps.bgs.ac.uk/geologyofbritain/home.html) Geology of Britain viewer (centred on South Marston).

Soil Survey of England and Wales, 1983. *Soils of England and Wales, Sheet 5 South West England*.

English Heritage, 2008. *Geophysical Survey in Archaeological Field Evaluation.*

Institute for Archaeologists. *Standard and Guidance for Archaeological Geophysical Survey.* http://www.archaeologists.net/sites/default/files/nodefiles/Geophysics2010.pdf

# APPENDIX A – METHODOLOGY & SURVEY EQUIPMENT

***Grid locations***

The location of the survey grids has been plotted together with the referencing information. Grids were set out using a Trimble R8 Real Time Kinematic (RTK) VRS Now GPS system and referenced to suitable topographic features around the perimeter of the site.

An RTK GPS can locate a point on the ground to a far greater accuracy than a standard GPS unit. A standard GPS suffers from errors created by satellite orbit errors, clock errors and atmospheric interference, resulting in an accuracy of 5m-10m. An RTK system uses a single base station receiver and a number of mobile units. The base station re-broadcasts the phase of the carrier it measured, and the mobile units compare their own phase measurements with those they received from the base station. A SmartNet RTK GPS uses Ordnance Survey's network of over 100 fixed base stations to give an accuracy of around 0.01m.

***Survey equipment and gradiometer configuration***

Although the changes in the magnetic field resulting from differing features in the soil are usually weak, changes as small as 0.2 nanoTeslas (nT) in an overall field strength of 48,000nT, can be accurately detected using an appropriate instrument.

The mapping of the anomaly in a systematic manner will allow an estimate of the type of material present beneath the surface. Strong magnetic anomalies will be generated by buried iron-based objects or by kilns or hearths. More subtle anomalies such as pits and ditches can be seen if they contain more humic material which is normally rich in magnetic iron oxides when compared with the subsoil.

To illustrate this point, the cutting and subsequent silting or backfilling of a ditch may result in a larger volume of weakly magnetic material being accumulated in the trench compared to the undisturbed subsoil. A weak magnetic anomaly should therefore appear in plan along the line of the ditch.

The magnetic survey was carried out using a dual sensor Grad601-2 Magnetic Gradiometer manufactured by Bartington Instruments Ltd. The instrument consists of two fluxgates very accurately aligned to nullify the effects of the Earth's magnetic field. Readings relate to the difference in localised magnetic anomalies compared with the general magnetic background. The Grad601-2 consists of two high stability fluxgate gradiometers suspended on a single frame. Each gradiometer has a 1m separation between the sensing elements so enhancing the response to weak anomalies.

***Sampling interval***

Readings were taken at 0.25m centres along traverses 1m apart. This equates to 3600 sampling points in a full 30m x 30m grid.

***Depth of scan and resolution***

The Grad 601-2 has a typical depth of penetration of 0.5m to 1.0m, though strongly magnetic objects may be visible at greater depths. The collection of data at 0.25m centres provides an optimum methodology for the task balancing cost and time with resolution.

***Data capture***

The readings are logged consecutively into the data logger which in turn is daily down- loaded into a portable computer whilst on site. At the end of each site survey, data is transferred to the office for processing and presentation.

# APPENDIX B – BASIC PRINCIPLES OF MAGNETIC SURVEY

Detailed magnetic survey can be used to effectively define areas of past human activity by mapping spatial variation and contrast in the magnetic properties of soil, subsoil and bedrock.

Weakly magnetic iron minerals are always present within the soil and areas of enhancement relate to increases in *magnetic susceptibility* and permanently magnetised *thermoremanent* material.

Magnetic susceptibility relates to the induced magnetism of a material when in the presence of a magnetic field. This magnetism can be considered as effectively permanent as it exists within the Earth's magnetic field. Magnetic susceptibility can become enhanced due to burning and complex biological or fermentation processes.

Thermoremanence is a permanent magnetism acquired by iron minerals that, after heating to a specific temperature known as the Curie Point, are effectively demagnetised followed by re-magnetisation by the Earth's magnetic field on cooling. Thermoremanent archaeological features can include hearths and kilns and material such as brick and tile may be magnetised through the same process.

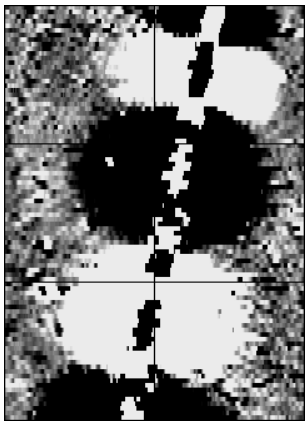
Silting and deliberate infilling of ditches and pits with magnetically enhanced soil creates a relative contrast against the much lower levels of magnetism within the subsoil into which the feature is cut. Systematic mapping of magnetic anomalies will produce linear and discrete areas of enhancement allowing assessment and characterisation of subsurface features. Material such as subsoil and non-magnetic bedrock used to create former earthworks and walls may be mapped as areas of lower enhancement compared to surrounding soils.

Magnetic survey is carried out using a fluxgate gradiometer which is a passive instrument consisting of two sensors mounted vertically 1m apart. The instrument is carried about 30cm above the ground surface and the top sensor measures the Earth's magnetic field whilst the lower sensor measures the same field but is also more affected by any localised buried field. The difference between the two sensors will relate to the strength of a magnetic field created by a buried feature, if no field is present the difference will be close to zero as the magnetic field measured by both sensors will be the same.

Factors affecting the magnetic survey may include soil type, local geology, previous human activity, disturbance from modern services etc.

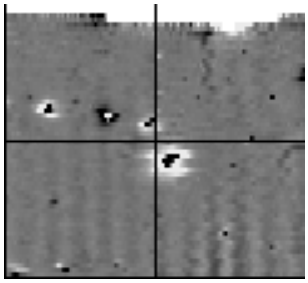
# APPENDIX C – GLOSSARY OF MAGNETIC ANOMALIES

**Bipolar**



A bipolar anomaly is one that is composed of both a positive response and a negative response. It can be made up of any number of positive responses and negative responses. For example a pipeline consisting of alternating positive and negative anomalies is said to be bipolar. See also dipolar which has only one area of each polarity. The interpretation of the anomaly will depend on the magnitude of the magnetic field strength. A weak response may be caused by a clay field drain while a strong response will probably be caused by a metallic service.

**Dipolar**

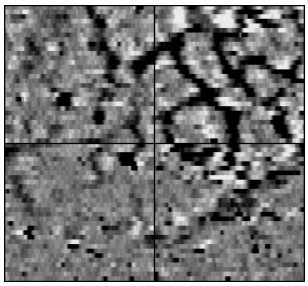


This consists of a single positive anomaly with an associated negative response. There should be no separation between the two polarities of response. These responses will be created by a single feature. The interpretation of the anomaly will depend on the magnitude of the magnetic measurements. A very strong anomaly is likely to be caused by a ferrous object.

**Positive anomaly with associated negative response**

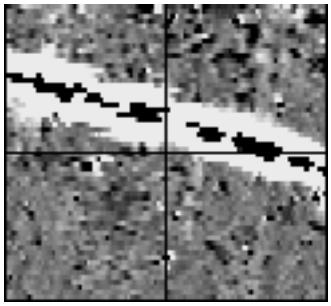
See bipolar and dipolar.

**Positive linear**



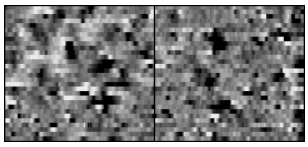
A linear response which is entirely positive in polarity. These are usually related to in-filled cut features where the fill material is magnetically enhanced compared to the surrounding matrix. They can be caused by ditches of an archaeological origin, but also former field boundaries, ploughing activity and some may even have a natural origin.

**Positive linear anomaly with associated negative response**



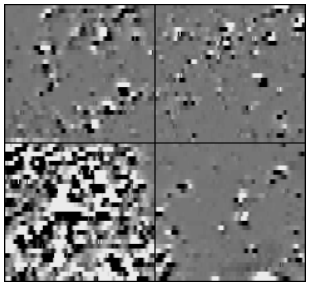
A positive linear anomaly which has a negative anomaly located adjacently. This will be caused by a single feature. In the example shown this is likely to be a single length of wire/cable probably relating to a modern service. Magnetically weaker responses may relate to earthwork style features and field boundaries.

**Positive point/area**



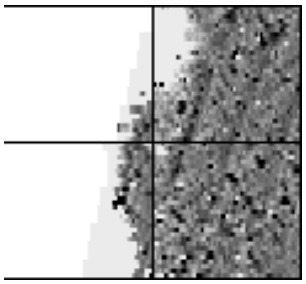
These are generally spatially small responses, perhaps covering just 3 or 4 reading nodes. They are entirely positive in polarity. Similar to positive linear anomalies they are generally caused by in-filled cut features. These include pits of an archaeological origin, possible tree bowls or other naturally occurring depressions in the ground.

**Magnetic debris**



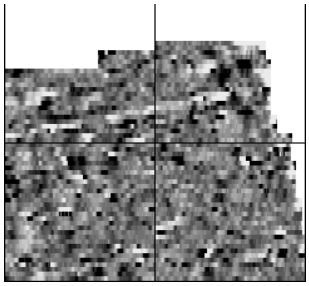
Magnetic debris consists of numerous dipolar responses spread over an area. If the amplitude of response is low (+/-3nT) then the origin is likely to represent general ground disturbance with no clear cause, it may be related to something as simple as an area of dug or mixed earth. A stronger anomaly (+/-250nT) is more indicative of a spread of ferrous debris. Moderately strong anomalies may be the result of a spread of thermoremanent material such as bricks or ash.

**Magnetic disturbance**



Magnetic disturbance is high amplitude and can be composed of either a bipolar anomaly, or a single polarity response. It is essentially associated with magnetic interference from modern ferrous structures such as fencing, vehicles or buildings, and as a result is commonly found around the perimeter of a site near to boundary fences.

**Negative linear**

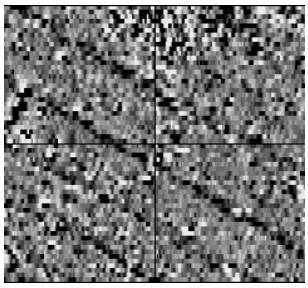


A linear response which is entirely negative in polarity. These are generally caused by earthen banks where material with a lower magnetic magnitude relative the background top soil is built up. See also ploughing activity.

**Negative point/area**

Opposite to positive point anomalies these responses may be caused by raised areas or earthen banks. These could be of an archaeological origin or may have a natural origin.

**Ploughing activity**



Ploughing activity can often be visualised by a series of parallel linear anomalies. These can be of either positive polarity or negative polarity depending on site specifics. It can be difficult to distinguish between ancient ploughing and more modern ploughing, clues such as the separation of each linear, straightness, strength of response and cross cutting relationships can be used to aid this, although none of these can be guaranteed to differentiate between different phases of activity.

**Polarity**

Term used to describe the measurement of the magnetic response. An anomaly can have a positive polarity (values above 0nT) and/or a negative polarity (values below 0nT).

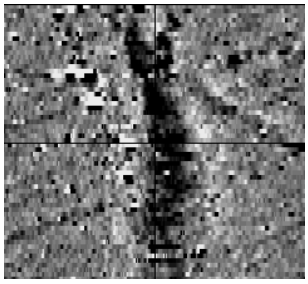
**Strength of response**

The amplitude of a magnetic response is an important factor in assigning an interpretation to a particular anomaly. For example a positive anomaly covering a $10m^2$ area may have values up to around 3000nT, in which case it is likely to be caused by modern magnetic interference. However, the same size and shaped anomaly but with values up to only 4nT may have a natural origin. Colour plots are used to show the amplitude of response.

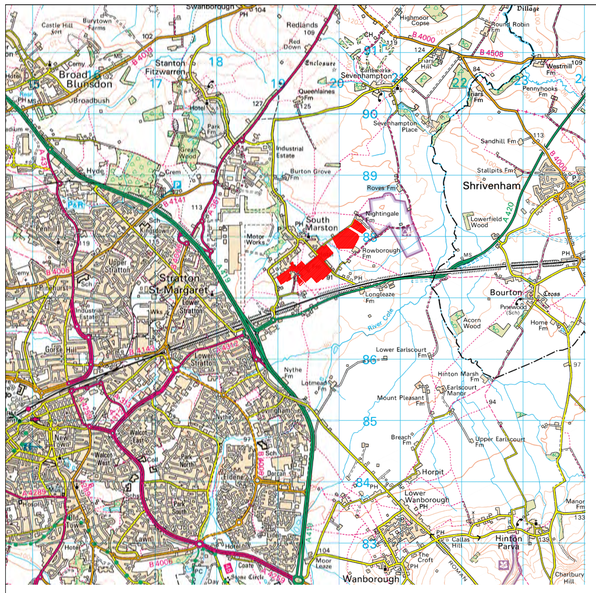
**Thermoremanent response**

A feature which has been subject to heat may result in it acquiring a magnetic field. This can be anything up to approximately +/-100 nT in value. These features include clay fired drains, brick, bonfires, kilns, hearths and even pottery. If the heat application has occurred in situ (e.g. a kiln) then the response is likely to be bipolar compared to if the heated objects have been disturbed and moved relative to each other, in which case they are more likely to take an irregular form and may display a debris style response (e.g. ash).

**Weak background variations**



Weakly magnetic wide scale variations within the data can sometimes be seen within sites. These usually have no specific structure but can often appear curvy and sinuous in form. They are likely to be the result of natural features, such as soil creep, dried up (or seasonal) streams. They can also be caused by changes in the underlying geology or soil type which may contain unpredictable distributions of magnetic minerals, and are usually apparent in several locations across a site.

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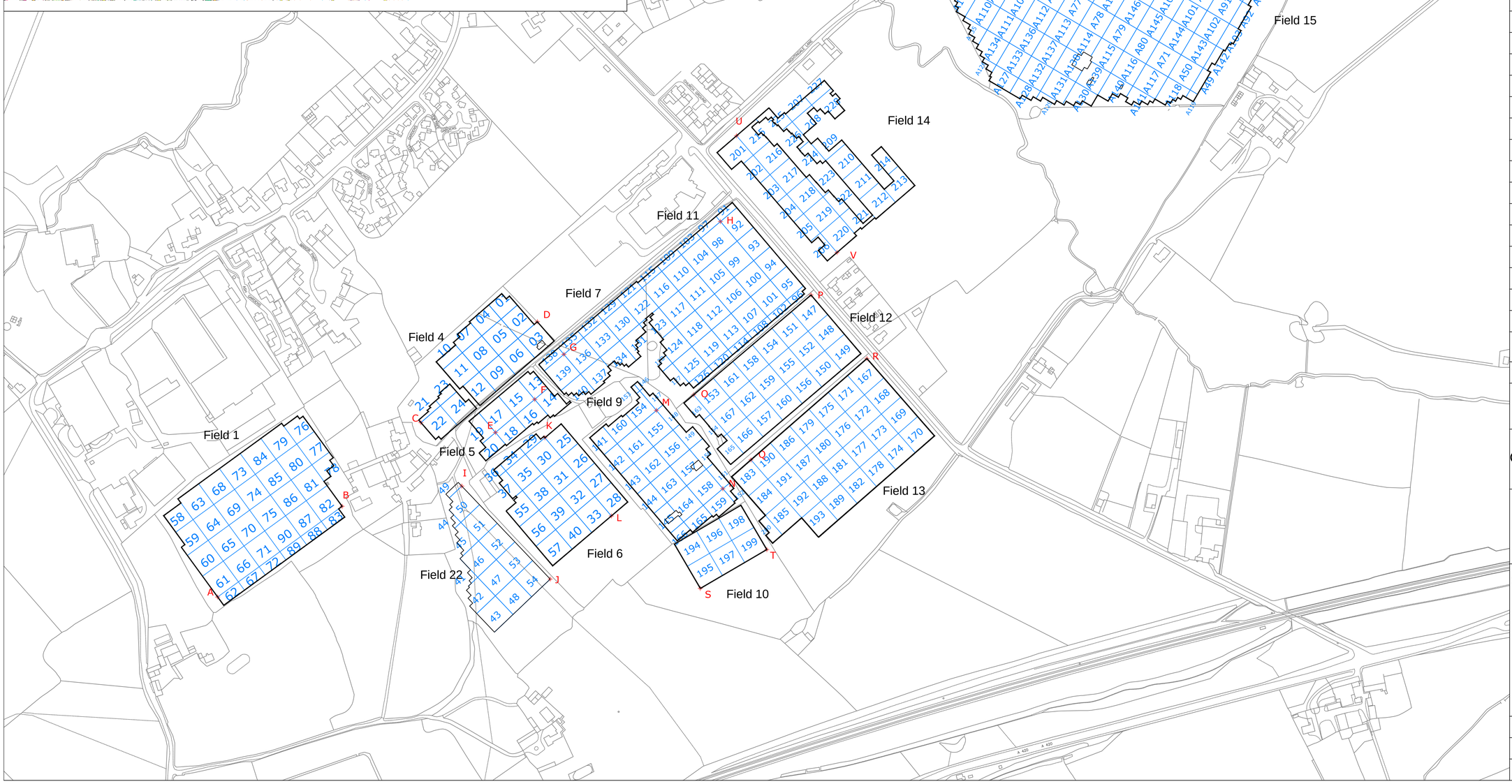
Vineyard House
Upper Hook Road
Upton Upon Severn
WR8 0SA

OS 100km square = SU

Site centred on NGR SU 200 870




## Amendments

| Issue No. | Date | Description |
|---|---|---|
| - | - | - |
| - | - | - |

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## KEY

| | |
|---|---|
| | Survey area |
| 12 | Numbered survey grid |
| A | GPS referencing point |

## OS REFERENCING INFORMATION

| | |
|---|---|
| A | 419048.596,187231.893 |
| B | 419194.095,187337.865 |
| C | 419286.217,187435.760 |
| D | 419422.252,187553.636 |
| E | 419373.288,187424.151 |
| F | 419419.149,187462.839 |
| G | 419453.256,187515.471 |
| H | 419636.260,187670.743 |
| I | 419333.936,187361.501 |
| J | 419437.289,187252.790 |
| K | 419431.097,187418.084 |
| L | 419508.746,187326.593 |
| M | 419561.386,187449.949 |
| N | 419639.010,187358.436 |
| O | 419604.965,187468.290 |
| P | 419742.116,187584.864 |
| Q | 419671.862,187392.147 |
| R | 419807.413,187510.579 |
| S | 419612.696,187241.902 |
| T | 419690.427,187287.266 |
| U | 419655.035,187770.979 |
| V | 419772.454,187634.551 |
| W | 420110.906,188112.132 |
| X | 420291.808,188005.481 |
| Y | 420218.393,188176.332 |
| Z | 420373.503,188085.003 |

| Job No. | Survey Date |
|---|---|
| J3305 | MARCH 13 |

Client

CGMS

Project Title

GEOPHYSICAL SURVEY - LAND EAST OF SWINDON

Subject

LOCATION OF SURVEY GRIDS & REFERENCING

STRATASCAN™
GEOPHYSICS FOR ARCHAEOLOGY AND ENGINEERING
VINEYARD HOUSE
UPTON UPON SEVERN
WR8 0SA
T: 01684 592266
E: info@stratascan.co.uk
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EUROPEAN GPR ASSOCIATION

SUMO GROUP MEMBER

Scale 1:2500 (0m 50 100 150m)

| Plot | Checked by | Issue No. |
|---|---|---|
| A1 | PPB | 01 |
| **Date** | **Drawn by** | **Figure No.** |
| APRIL 13 | ELW | 01 |